PRODUCT USER MANUAL

For OMI_CIRCULATION_VOLTRANS_ARCTIC

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TABLE OF CONTENTS

RECORD TABLE **2**

GLOSSARY AND ABBREVIATIONS **4**

DATA ACCESS **5**

1) INTRODUCTION **6**

a) Summary 6

2) DESCRIPTION OF THE PRODUCT SPECIFICATION **7**

a) General Information 7

b) Production System Description 7

c) Dataset details 8

3) FILE FORMAT **9**

4) FILES NOMENCLATURE **9**

5) REFERENCES **10**

GLOSSARY AND ABBREVIATIONS

NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_CIRCULATION_VOLTRANS_ARCTIC_averaged. It provides time series of Atlantic Water volume transport into the Nordic Seas via the Faroe-Shetland Channel and into the Barents Sea via the Barents Sea Opening, as well as into the Polar Basin via the Fram Strait. In addition, the transports of overflow waters out from the Nordic Seas via the Faroe-Shetland Channel and the Denmark Strait are also provided. The OMI contains five time series:

- VOL_FSCAW contain monthly means of Atlantic Water volume transport into the Nordic Seas via the Faroe-Shetland Channel.
- VOL_BSOAW contain monthly means of Atlantic Water volume transport into the Barents Sea.
- VOL_FSAW contain monthly means of Atlantic Water volume transport into the Polar Basin via the Fram Strait.
- VOL_FSCOW contain monthly means of overflow waters exiting the Nordic Seas via the Faroe-Shetland Channel.
- VOL_DSOW contain monthly means of overflow waters exiting the Nordic Seas via the Denmark Strait.

The time series are computed using the TOPAZ5 reanalysis from the Arctic Marine Forecasting Centre (product name ARCTIC_REANALYSIS_PHY_002_003).
The product is published after automatic and delayed-mode quality controls and is available online and disseminated through the Copernicus Marine Service Marine Data Store. Files downloaded are in NetCDF format and follow the CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

Product OMI is described in the Quality Information Document (QUID) CMEMS-OMI-QUID-ARC-CIRCULATION_VOLTRANS_averaged.pdf (See References).

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_CIRCULATION_VOLTRANS_ARCTIC_averaged
Geographical coverage	Faroe Shetland Channel (61°N 20' – 59°N 30'; 6°W 30' - 3°W) Barents Sea Opening (71°N 30' - 73°N 30'; 20°E - 19°E 30') Fram Strait (78°N 30' - 78°N 52'; 7°W 12' - 9°E 2') Denmark Strait (66°N 22' - 66°N 15'; 37°W - 22°W 40')
Variables	VOL_FSCAW VOL_BSOAW VOL_FSAW VOL_DSOW VOL_FSCOW
Available time series	1991-01-15 to 2024-12-15
Temporal resolution	Monthly means
Horizontal resolution	N/A
Integration depth (for example)	Surface to bottom
Water mass criterion	Faroe-Shetland Channel Atlantic Water: T > 5 °C Barents Sea Opening Atlantic Water: T > 3 °C Fram Strait Atlantic Water: T > 2 °C Faroe-Shetland Channel Overflow Water: σ_θ > 27.8 kg m^{-3} Denmark Strait Overflow Water; σ_θ > 27.8 kg m^{-3}
Format	NetCDF4

b) Production System Description

The estimates are derived from the Arctic reanalysis and forecasting products ARCTIC_MULTIYEAR_PHY_002_003 and ARCTIC_ANALYSISFORECAST_PHY_002_001 (see References for documentation).

This product provides the modelled ocean volume transport of Atlantic Water into the Nordic Seas via the Faroe-Shetland Channel (T > 5 °C); into the Barents Sea via the Barents Sea Opening (T > 3 °C); and into the Polar Basin via the Fram Strait (T > 2 °C). In addition, the modelled ocean volume transport of overflow waters exiting the Nordic

Seas via the Faroe-Shetland Channel and the Denmark Strait (σ_θ > 27.8 kg m^{-3}) is also provided (see Ch. 3.2 of the Ocean State Report #3 (von Schuckmann et al., 2018) for details).

c) Dataset details

OMI_CIRCULATION_VOLTRANS_ARCTIC_averaged contains the monthly averaged volume transports of Atlantic Water and overflow water entering and exiting the Nordic Seas.
Vol_FSCAW_Reanalysis [10^6 m^{3}s^{-1}] Atlantic Water volume transport through the Faroe-Shetland Channel (from reanalysis) ocean_volume_transport_across_line **Vol_BSOAW_Reanalysis [10^6 m^{3}s^{-1}]** Atlantic Water volume transport through the Barents Sea Opening (from reanalysis) ocean_volume_transport_across_line **Vol_FSAW_Reanalysis [10^6 m^{3}s^{-1}]** Atlantic Water volume transport through the Fram Strait (from reanalysis) ocean_volume_transport_across_line **Vol_DSOW_Reanalysis [10^6 m^{3}s^{-1}]** Overflow water volume transport through the Denmark Strait (from reanalysis) ocean_volume_transport_across_line **Vol_FSCOW_Reanalysis [10^6 m^{3}s^{-1}]** Overflow water volume transport through the Faroe-Shetland Channel (from reanalysis) ocean_volume_transport_across_line

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

5) REFERENCES

QUID : https://documentation.marine.copernicus.eu/QUID/CMEMS-ARC-QUID-CIRCULATION-VOLTRANS.pdf

https://documentation.marine.copernicus.eu/PUM/CMEMS-ARC-PUM-002-ALL.pdf

https://documentation.marine.copernicus.eu/QUID/CMEMS-ARC-QUID-002-001.pdf

v. Schuckmann K, et al. 2018. *Copernicus Marine Service Ocean State Report,* Journal of Operational Oceanography, 11:sup1, S1-S142.